# Global Aviation Industry 2010-2026

## About Dataset

### Global Aviation Industry 2010-2026

**Subtitle:** 30 airlines × 17 years financials, Boeing vs Airbus order book, regional passenger traffic, 40 top routes, and 40 major aviation incidents — covering the 737 MAX crisis, COVID devastation, Saudi aviation push, and 2024-2025 Boeing crisis

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### Tags

aviation · airlines · boeing · airbus · transportation · tourism · aerospace · time series · economics · tabular

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### Overview

A comprehensive dataset tracking the **global aviation industry from 2010 to 2026** — covering Boeing's 737 MAX crisis ($150B+ market cap loss), Airbus's market share gain, COMAC C919 emergence, COVID's -86% Asia Pacific traffic collapse, the post-pandemic recovery, Saudi Arabia's aviation mega-projects (Riyadh Air), and the 2024-2025 Boeing crisis (Alaska 1282 door plug, FAA cap on production).

This dataset is designed for **aviation analysts, equity analysts (airline/aerospace sector), travel industry researchers, transportation policy researchers, and ML practitioners working on aviation forecasting**.

All anchor data sourced from public materials: airline 10-K filings (SEC EDGAR), Boeing/Airbus monthly orders & deliveries reports, IATA Air Passenger Market Analysis reports, ASN Aviation Safety Network database, ICAO traffic statistics.

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### Files

| File | Rows | Description |
|---|---|---|
| airline_financials.csv | 497 | 30 airlines × 17 years: revenue, margins, passengers, load factor, fleet |
| fleet_orders.csv | 86 | Boeing vs Airbus vs COMAC orders/deliveries/backlog by family |
| passenger_traffic.csv | 1,176 | 6 regions × monthly RPK/ASK/load factor 2010-2026 |
| route_performance.csv | 400 | 40 top global routes × multi-year passengers, fares, revenue |
| aviation_incidents.csv | 40 | Major accidents and incidents 2010-2026 |

Total: ~2,200 rows, ~150 KB.

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**30 Airlines Covered**

**US Legacy (4):** Delta, United, American, Alaska **US Low-Cost (2):** Southwest, JetBlue **European Legacy (4):** Lufthansa Group, IAG, Air France-KLM, Aeroflot **European Low-Cost (2):** Ryanair, easyJet **European Hybrid (1):** Turkish Airlines **Chinese (3):** Air China, China Eastern, China Southern **Japanese (2):** JAL, ANA **Other Asia (5):** Singapore Airlines, Cathay Pacific, Korean Air, IndiGo, AirAsia **Middle East (5):** Emirates, Qatar Airways, Etihad, Saudia, Riyadh Air **Latin America (2):** LATAM, Azul

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**Key Findings Captured**

**COVID Devastation 2020**

- **Delta**: $47B (2019) → $17B (2020) = -64%, operating margin -36%
- **Lufthansa Group**: $40B → $15B
- **Cathay Pacific** worst affected: revenue -58%, 4 years to recover
- **Asia Pacific RPK collapse**: 64B (Dec 2019) → 9B (April 2020) = -86%
- Industry-wide operating margins -30% to -35% in 2020

**Boeing 737 MAX Crisis**

- **2018**: Boeing 737 family net orders +675
- **2019**: Net orders **-87** (cancellations exceeded orders post-ET302 grounding)
- **2020**: Net orders **-1,026** (catastrophic, MAX still grounded + COVID)
- **2024**: Net orders +569 vs Airbus A320 +1,456 (2.5× gap)
- Recovery still incomplete in 2026

**2024-2025 Boeing Crisis**

- **AS1282** (Jan 2024): Alaska 737 MAX 9 door plug blowout
- **JL516** (Jan 2024): JAL A350 collided with Coast Guard at Haneda
- **AA5342** (Jan 2025): American Eagle CRJ-700 mid-air collision with Army Black Hawk near DCA
- **DL4819** (Feb 2025): Delta CRJ-900 flipped on landing in Toronto, all survived
- **AI171** (June 2025): Air India 787 crash in Ahmedabad
- Boeing production cap at 38/month by FAA, recovery to 50/month delayed to 2026

**COMAC C919 Rise**

- **2018**: 24 orders, 0 deliveries
- **2023**: 250 orders, first 4 deliveries
- **2024**: 220 orders, 13 deliveries
- China's domestic narrowbody program finally ramping
- ~80% domestic Chinese orders; export market still limited by certification

**Middle East Growth**

- **Emirates 2024**: $33B revenue (largest international airline by revenue)

- **Qatar Airways**: $7B (2010) → $22B (2024)
- **Saudia + Riyadh Air**: Saudi Arabia investing in becoming aviation hub
- **Etihad** restructured: $9B peak (2015) → $7B (2024)

**Recovery Asymmetry**

- **North America recovery to 2019 baseline**: 2023
- **Europe recovery to 2019**: 2024
- **Asia Pacific recovery**: 2024-2025 (China lockdowns delayed)
- Middle East: exceeded 2019 by **2022**

**Geopolitical Aviation Events**

- **MH17** (2014): Shot down over Ukraine, 298 fatalities
- **PS752** (2020): Shot down by IRGC, Tehran
- **FR4978** (2021): Ryanair forced down by Belarus, Roman Protasevich arrested
- **Aeroflot 2022+**: Sanctions cut Russian airlines from Western MRO services
- **J28243** (Dec 2024): Azerbaijan Airlines hit by Russian air defense

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**Column Descriptions**

**airline_financials.csv**

| Column | Description |
| --- | --- |
| year, airline_name, iata_code, country_iso3, region | Identifiers |
| business_model | legacy / low_cost / regional |
| revenue_usd_bn | Annual revenue (USD billions) |
| operating_margin_pct | Operating margin (negative in 2020-2021) |
| operating_income_usd_bn | Revenue × margin |

| Column | Description |
| --- | --- |
| passengers_carried_m | Passengers in millions |
| load_factor_pct | Passenger load factor |
| fleet_size_est | Approximate fleet size |

**fleet_orders.csv**

| Column | Description |
| --- | --- |
| year, manufacturer, aircraft_family | Boeing/Airbus/COMAC/Embraer |
| orders_gross, orders_net | Gross and net (after cancellations) orders |
| deliveries | Aircraft delivered that year |
| backlog_end_of_year | Outstanding orders at year-end |
| is_widebody, is_narrowbody, is_regional | Boolean classification |

**passenger_traffic.csv**

| Column | Description |
| --- | --- |
| year_month, year, month | Monthly time index |
| region | North America / Europe / Asia Pacific / Middle East / Latin America / Africa |
| rpk_billions | Revenue Passenger Kilometers (industry standard demand metric) |
| ask_billions | Available Seat Kilometers (supply metric) |
| load_factor_pct | RPK / ASK ratio |

**route_performance.csv**

| Column | Description |
| --- | --- |
| year, route, origin_iata, destination_iata | Identifiers |
| distance_km | Great circle distance |
| region, main_airlines | Context |
| annual_passengers_m | Annual passengers in millions |

| Column | Description |
|---|---|
| avg_fare_usd | Average one-way fare |
| weekly_frequency_est | Flights per week |
| annual_revenue_usd_m | Estimated annual route revenue |

**aviation_incidents.csv**

| Column | Description |
|---|---|
| incident_id, date, year, month | Identifiers |
| flight_number, airline, aircraft_type | Flight details |
| severity | fatal_crash / fatal_collision / incident / ground_incident / sanctions_impact |
| fatalities | Death count |
| location, description | Plain-language details |
| is_fatal, is_geopolitical, is_boeing, is_airbus | Boolean flags |

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**Source**

**Real public data:**

- **Airline financials**: SEC 10-K filings for US airlines (Delta, United, American, Southwest, Alaska, JetBlue), foreign equivalents for IAG, Lufthansa, Air France-KLM, Ryanair, easyJet, Turkish, Aeroflot, JAL, ANA, Singapore Airlines, Cathay Pacific
- **Boeing/Airbus orders**: Monthly orders & deliveries reports (publicly posted by both manufacturers)
- **Passenger traffic**: IATA Air Passenger Market Analysis monthly reports, ICAO traffic statistics
- **Top routes**: OAG, IATA route statistics, airline schedule data
- **Aviation incidents**: ASN Aviation Safety Network database, NTSB reports, EASA/FAA notices

**Modeled components:**

- Year-by-year revenue interpolation between 10-K anchor points
- Operating margins calibrated with COVID/MAX/Russia sanctions adjustments
- Regional passenger traffic generated from real anchor years with seasonal patterns
- Route passenger volumes derived from aggregate regional data and distance models
- Specific incidents are all real and verified; flight numbers, dates, fatalities accurate

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**Methodology**

**Step 1 — Airline Financials**

For each airline, anchored to actual 10-K revenue at 8-10 reference years. Linear interpolation with 3% noise. Operating margins use segment norms (legacy 6-8%, low-cost 12%, regional 8%) with crisis adjustments: COVID 2020 -35%, 2021 -18%, Boeing 737 MAX effect on US carriers 2019, Aeroflot sanctions 2022+, Saudi investment phase margins (Riyadh Air -20% during build-out).

**Step 2 — Fleet Orders**

Real anchor data from Boeing/Airbus monthly reports at 8 key years (2010, 2013, 2015, 2017, 2018, 2019, 2020, 2021, 2022, 2023, 2024). Linear interpolation between anchors for non-anchor years. COMAC and Embraer separately tracked.

**Step 3 — Passenger Traffic**

Regional RPK anchored to IATA monthly reports at 7 reference years. Monthly interpolation with seasonal pattern (summer peak Aug, winter trough Feb for Northern hemisphere; opposite for Latin America). COVID dynamics specifically modeled: April-June 2020 = -85%, H2 2020 = -65%, 2021 = -55%, H1 2022 = -35%, recovery to 2019 baseline by 2023-2025 depending on region.

**Step 4 — Route Performance**

40 real top global routes with verified distances and dominant airlines. Per-route passenger volumes calibrated to distance (short-haul = higher volume, long-haul = lower volume). Year-over-year factors apply COVID dynamics. Avg fare scales with distance plus year-over-year inflation.

**Step 5 — Aviation Incidents**

40 real verified incidents from ASN/NTSB/EASA databases. All fatalities, dates, flight numbers, locations accurate. Categorized as fatal_crash, fatal_collision, incident, ground_incident, or sanctions_impact.

**Reproducibility**

Deterministic given seed. Source code generate_aviation.py included.

**What This Dataset Is and Is Not**

**Suitable for:** aviation industry research, equity analysis (airline/aerospace sector), trend analysis, COVID recovery analysis, Boeing vs Airbus competitive analysis, route economics modeling, ML training for aviation forecasting, journalism.

**Not suitable for:** individual incident audit (use ASN/NTSB primary data), regulatory compliance, specific airline financial verification (cross-reference 10-K filings), passenger-level analysis, individual flight scheduling.

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**Suggested Tasks**

**1. Boeing Recovery Modeling**

Boeing 737 MAX net orders went from +1,208 (2013) to -1,026 (2020) to +569 (2024). Build a recovery trajectory model. When does Boeing return to parity with Airbus on narrowbody?

**2. COVID Recovery Asymmetry**

Compare time to full recovery (2019 baseline) by region. Asia Pacific lagged 2 years behind. Quantify the lockdown effect.

**3. Saudi Aviation Strategy Analysis**

Saudia, Riyadh Air, and Etihad combined revenue trajectory 2023-2026. How much state investment is required to compete with Emirates/Qatar?

**4. Route Economics**

Average fare × passengers = route revenue. Build a route profitability model accounting for fuel/operating costs. Identify gold routes vs commodity routes.

**5. Low-Cost vs Legacy Margins**

Compare operating margins by business model over time. Did COVID change the structural margin gap?

**6. Incident Frequency Analysis**

40 major incidents in 16 years. Has commercial aviation become safer? Adjust for traffic volume.

**7. Boeing Crisis Impact on Stocks**

Cross-reference 2024 incidents (AS1282, JL516, AA5342, AI171) with Boeing stock data. What's the cumulative drag?

**8. Middle East Hub Battle**

Emirates, Qatar Airways, Turkish Airlines compete for global connecting traffic. Build a market share model.

**9. COMAC C919 Trajectory**

Project COMAC orders/deliveries 2026-2030 based on 2023-2025 ramp data. Realistic threat to Boeing/Airbus duopoly?

**10. Cross-Series Analysis**

Combine with companion datasets:

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**Limitations**

- **Foreign airline financials**: USD-equivalent revenue may differ slightly from local currency reported numbers due to FX
- **Chinese airlines 2020-2022**: Reporting transparency limited during COVID; estimates used
- **Fleet orders for 2025-2026**: Projections based on H1 2026 trajectory
- **Route passenger volumes**: Based on regional aggregates and route prominence, not OAG schedule-level granularity
- **Incidents**: Only major incidents covered; full ASN database is much larger
- **No cargo airlines** (FedEx, UPS, Cargolux excluded)
- **No private jets / general aviation** included
- **End of period 2026 partial**: data through April 2026

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**Step 1 – Business Understanding & Data Dictionary**

**Data Dictionary:**

**Table 1: airline_financials**

| Column Name | Data Type | Description | Example |
|---|---|---|---|
| year | INTEGER | Financial reporting year | 2024 |
| airline_name | VARCHAR(100) | Full airline name | Delta Air Lines |
| iata_code | VARCHAR(5) | IATA airline code | DL |
| country_iso3 | CHAR(3) | ISO-3 country code | USA |
| region | VARCHAR(50) | Geographic region | North America |
| business_model | VARCHAR(20) | Airline type (Legacy, Low Cost, Hybrid) | Legacy |
| revenue_usd_bn | DECIMAL(10,2) | Annual revenue in USD billions | 58.20 |
| operating_margin_pct | DECIMAL(5,2) | Operating profit margin (%) | 9.50 |
| operating_income_usd_bn | DECIMAL(10,2) | Operating income in USD billions | 5.53 |
| passengers_carried_m | DECIMAL(10,2) | Passengers carried (millions) | 190.30 |
| load_factor_pct | DECIMAL(5,2) | Passenger load factor (%) | 86.40 |
| fleet_size_est | INTEGER | Estimated fleet size | 982 |

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**Table 2: fleet_orders**

| Column Name | Data Type | Description | Example |
|---|---|---|---|
| year | INTEGER | Calendar year | 2024 |
| manufacturer | VARCHAR(30) | Aircraft manufacturer | Boeing |
| aircraft_family | VARCHAR(50) | Aircraft family | 737 MAX |
| orders_gross | INTEGER | Total aircraft orders | 623 |
| orders_net | INTEGER | Net orders after cancellations | 569 |
| deliveries | INTEGER | Aircraft delivered | 348 |
| backlog_end_of_year | INTEGER | Outstanding orders | 5124 |
| is_widebody | BOOLEAN | Widebody aircraft indicator | FALSE |
| is_narrowbody | BOOLEAN | Narrowbody aircraft indicator | TRUE |
| is_regional | BOOLEAN | Regional aircraft indicator | FALSE |

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**Table 3: passenger_traffic**

| Column Name | Data Type | Description | Example |
|---|---|---|---|
| year_month | DATE | Year and month | 2024-07 |
| year | INTEGER | Calendar year | 2024 |

| Column Name | Data Type | Description | Example |
|---|---|---|---|
| month | INTEGER | Month number | 7 |
| region | VARCHAR(50) | Geographic region | Asia Pacific |
| rpk_billions | DECIMAL(10,2) | Revenue Passenger Kilometers (billions) | 61.40 |
| ask_billions | DECIMAL(10,2) | Available Seat Kilometers (billions) | 72.10 |
| load_factor_pct | DECIMAL(5,2) | Passenger load factor (%) | 85.20 |

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**Table 4: route_performance**

| Column Name | Data Type | Description | Example |
|---|---|---|---|
| year | INTEGER | Calendar year | 2024 |
| route | VARCHAR(100) | Flight route | LHR-JFK |
| origin_iata | CHAR(3) | Origin airport code | LHR |
| destination_iata | CHAR(3) | Destination airport code | JFK |
| distance_km | INTEGER | Route distance in kilometers | 5540 |
| region | VARCHAR(50) | Route region | Europe–North America |
| main_airlines | VARCHAR(255) | Major airlines operating the route | British Airways, American Airlines |
| annual_passengers_m | DECIMAL(10,2) | Annual passengers (millions) | 4.80 |
| avg_fare_usd | DECIMAL(10,2) | Average one-way fare (USD) | 780.00 |
| weekly_frequency_est | INTEGER | Estimated weekly flights | 140 |
| annual_revenue_usd_m | DECIMAL(12,2) | Annual route revenue (USD millions) | 3744.00 |

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**Table 5: aviation_incidents**

| Column Name | Data Type | Description | Example |
|---|---|---|---|
| incident_id | VARCHAR(20) | Unique incident identifier | AI171 |
| date | DATE | Incident date | 2025-06-12 |
| year | INTEGER | Calendar year | 2025 |
| month | INTEGER | Month number | 6 |
| flight_number | VARCHAR(20) | Flight number | AI171 |
| airline | VARCHAR(100) | Airline involved | Air India |
| aircraft_type | VARCHAR(100) | Aircraft model | Boeing 787-8 |
| severity | VARCHAR(30) | Incident severity | fatal_crash |
| fatalities | INTEGER | Number of fatalities | 241 |

| Column Name | Data Type | Description | Example |
|---|---|---|---|
| location | VARCHAR(150) | Incident location | Ahmedabad, India |
| description | TEXT | Incident summary | Aircraft crashed shortly after takeoff |
| is_fatal | BOOLEAN | Fatal incident flag | TRUE |
| is_geopolitical | BOOLEAN | Geopolitical event flag | FALSE |
| is_boeing | BOOLEAN | Boeing aircraft flag | TRUE |
| is_airbus | BOOLEAN | Airbus aircraft flag | FALSE |

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**Suggested Primary Keys**

| Table | Primary Key |
|---|---|
| airline_financials | (airline_name, year) |
| fleet_orders | (manufacturer, aircraft_family, year) |
| passenger_traffic | (year_month, region) |
| route_performance | (route, year) |
| aviation_incidents | incident_id |

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**Potential Relationships**

| From Table | Column | To Table | Column | Relationship |
|---|---|---|---|---|
| airline_financials | airline_name | aviation_incidents | airline | Airline-based analysis |
| airline_financials | year | fleet_orders | year | Year-based analysis |
| airline_financials | year | passenger_traffic | year | Time analysis |
| passenger_traffic | region | route_performance | region | Regional analysis |
| route_performance | year | fleet_orders | year | Year-based comparison |

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**Relationship Diagram (Business View)**

```
┌─────────────────────────────┐
│   airline_financials    │
├─────────────────────────────┤
│ year             │
│ airline_name         │
│ region             │
│ business_model         │
│ revenue             │
│ passengers           │
```

```
│ load_factor          │
└──────────────┬──────────────┘
       │
airline_name  │  year
       │
       ▼
┌───────────────────────────────┐
│ aviation_incidents     │
├───────────────────────────────┤
│ incident_id          │
│ airline              │
│ year                 │
│ aircraft_type        │
│ fatalities           │
└───────────────────────────────┘


┌───────────────────────────────┐
│ passenger_traffic     │
├───────────────────────────────┤
│ year_month           │
│ year                 │
│ region               │
│ RPK                  │
│ ASK                  │
│ Load Factor          │
└──────────────┬──────────────┘
       │
  region  │  year
       │
       ▼
┌───────────────────────────────┐
│ route_performance     │
├───────────────────────────────┤
│ route                │
│ region               │
│ year                 │
```

```
│ passengers        │
│ revenue           │
└───────────────────┘


┌───────────────────┐
│   fleet_orders    │
├───────────────────┤
│ year              │
│ manufacturer      │
│ aircraft_family   │
│ orders            │
│ deliveries        │
└───────────────────┘
```

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==================================================================**Step 2 – Data Profiling & Data Quality Assessment**

**Data Profiling Process**

Each dataset was reviewed using Microsoft Excel to understand its structure and evaluate data quality before importing it into PostgreSQL.

The following checks were performed:

- Dataset dimensions
- Column names
- Data types
- Missing values
- Duplicate records
- Text consistency
- Numeric validation
- Date validation
- Business rule validation
- Outlier identification

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**Data Quality Checklist**

| Validation Item | Purpose | Status |
| --- | --- | --- |
| Row Count Verification | Verify expected number of records | Completed |
| Column Count Verification | Ensure dataset structure is correct | Completed |

| | | |
|---|---|---|
| Column Name Review | Identify blank or inconsistent names | Completed |
| Data Type Validation | Verify numeric, text, and date columns | Completed |
| Missing Value Assessment | Detect incomplete information | Completed |
| Duplicate Record Check | Identify duplicate observations | Completed |
| Text Quality Check | Detect spaces and inconsistent text | Completed |
| Numeric Validation | Verify valid numeric ranges | Completed |
| Date Validation | Ensure valid date formats | Completed |
| Business Rule Validation | Identify logically invalid values | Completed |

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**Dataset Profiling Summary**

**1. airline_financials.csv**

**Columns Reviewed**

- year
- airline_name
- iata_code
- country_iso3
- region
- business_model
- revenue_usd_bn
- operating_margin_pct
- operating_income_usd_bn
- passengers_carried_m
- load_factor_pct
- fleet_size_est

**Validation Performed**

- Verified year values
- Checked airline names for consistency
- Reviewed revenue values
- Validated operating margins
- Verified passenger counts
- Checked fleet sizes
- Confirmed load factors remained within expected business limits

Potential issues identified during profiling were documented for further review before SQL analysis.

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**2. fleet_orders.csv**

**Columns Reviewed**

- year
- manufacturer
- aircraft_family
- orders_gross
- orders_net
- deliveries
- backlog_end_of_year
- is_widebody
- is_narrowbody
- is_regional

Validation included:

- Manufacturer consistency
- Aircraft family naming
- Boolean value verification
- Negative order detection
- Delivery validation
- Backlog validation

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**3. passenger_traffic.csv**

**Columns Reviewed**

- year_month
- year
- month
- region
- rpk_billions
- ask_billions
- load_factor_pct

Validation included:

- Monthly continuity
- Region consistency
- RPK values
- ASK values
- Load factor validation

- Date formatting

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**4. route_performance.csv**

**Columns Reviewed**

- year
- route
- origin_iata
- destination_iata
- distance_km
- region
- main_airlines
- annual_passengers_m
- avg_fare_usd
- weekly_frequency_est
- annual_revenue_usd_m

Validation included:

- Airport code consistency
- Route naming
- Distance validation
- Passenger volume review
- Fare validation
- Revenue validation

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**5. aviation_incidents.csv**

**Columns Reviewed**

- incident_id
- date
- year
- month
- flight_number
- airline
- aircraft_type
- severity
- fatalities
- location

- description
- is_fatal
- is_geopolitical
- is_boeing
- is_airbus

Validation included:

- Incident ID uniqueness
- Date consistency
- Airline naming
- Fatality counts
- Boolean value verification
- Severity classification

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**Data Quality Findings**

The following data quality dimensions were assessed throughout the profiling process.

**Missing Values**

Each column was reviewed to identify null or blank values that could impact future analysis.

Recommended action:

- Investigate missing business-critical fields.
- Decide whether to retain, replace, or remove incomplete records.

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**Duplicate Records**

Duplicate rows were reviewed to determine whether they represented true duplicates or valid repeated business events.

Recommended action:

- Remove only confirmed duplicate records.
- Preserve legitimate repeated observations.

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**Text Quality**

Text columns were inspected for:

- Leading spaces
- Trailing spaces
- Extra spaces
- Mixed capitalization
- Inconsistent naming conventions

Recommended action:

- Standardize formatting before importing into PostgreSQL.

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**Numeric Validation**

Numeric fields were reviewed for:

- Negative values
- Impossible percentages
- Invalid passenger counts
- Invalid fleet sizes
- Incorrect revenue values

Recommended action:

- Investigate values that violate business rules.

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**Date Validation**

Date fields were reviewed to ensure:

- Valid calendar dates
- Consistent formatting
- No missing values

Recommended format:

YYYY-MM-DD

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**Business Rule Validation**

The following business rules were considered during profiling:

- Passenger counts cannot be negative.
- Fleet size must be positive.
- Load factor should remain within realistic operating limits.
- Revenue should not be negative.
- Aircraft deliveries cannot be negative.

Records violating business rules were flagged for further investigation.

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**Outlier Assessment**

Extreme values were identified for:

- Revenue
- Passenger counts
- Average fares
- Fleet size

- Operating margins Outliers were reviewed to determine whether they represented genuine business events, such as the COVID-19 pandemic or major industry disruptions, rather than data entry errors.

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**Data Quality Assessment**

Overall, the datasets demonstrate a well-structured format suitable for analytical processing.

The profiling process confirmed that the data is appropriate for import into PostgreSQL after addressing any identified quality issues.

This assessment provides confidence that subsequent SQL analysis and Power BI reporting will be based on reliable and consistent data.

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**Step 3 – PostgreSQL Database Design & Data Import Documentation**

**Project Objective**

The objective of this phase is to design a structured PostgreSQL database for the Global Aviation Industry dataset and import all CSV files into the database. A properly designed database ensures data integrity, supports efficient SQL querying, and provides a reliable data source for Power BI dashboards.

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**Database Information**

**Database Name**

aviation_project

**Database Management System**

- PostgreSQL

**Database Administration Tool**

- pgAdmin 4

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**Dataset Overview**

The project consists of five datasets covering different aspects of the global aviation industry between 2010 and 2026.

| **Table** | **Description** | **Expected Rows** |
|---|---|---|
| airline_financials | Airline financial performance | 497 |
| fleet_orders | Aircraft manufacturer orders and deliveries | 86 |
| passenger_traffic | Monthly regional passenger traffic | 1,176 |
| route_performance | Global route performance | 400 |
| aviation_incidents | Major aviation incidents | 40 |

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**Database Schema**

The PostgreSQL database contains the following tables:

aviation_project

|

├── airline_financials

├── fleet_orders

├── passenger_traffic

├── route_performance

└── aviation_incidents

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**Table Structure**

**airline_financials**

| Column | Data Type |
|---|---|
| year | INTEGER |
| airline_name | VARCHAR(100) |
| iata_code | VARCHAR(5) |
| country_iso3 | CHAR(3) |
| region | VARCHAR(50) |
| business_model | VARCHAR(30) |
| revenue_usd_bn | NUMERIC(10,2) |
| operating_margin_pct | NUMERIC(5,2) |
| operating_income_usd_bn | NUMERIC(10,2) |
| passengers_carried_m | NUMERIC(10,2) |
| load_factor_pct | NUMERIC(5,2) |
| fleet_size_est | INTEGER |

Primary Key

(airline_name, year)

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**fleet_orders**

| Column | Data Type |
|---|---|
| year | INTEGER |
| manufacturer | VARCHAR(30) |
| aircraft_family | VARCHAR(50) |
| orders_gross | INTEGER |
| orders_net | INTEGER |
| deliveries | INTEGER |
| backlog_end_of_year | INTEGER |

| Column | Data Type |
|---|---|
| is_widebody | BOOLEAN |
| is_narrowbody | BOOLEAN |
| is_regional | BOOLEAN |

Primary Key

(manufacturer, aircraft_family, year)

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**passenger_traffic**

| Column | Data Type |
|---|---|
| year_month | DATE |
| year | INTEGER |
| month | INTEGER |
| region | VARCHAR(50) |
| rpk_billions | NUMERIC(10,2) |
| ask_billions | NUMERIC(10,2) |
| load_factor_pct | NUMERIC(5,2) |

Primary Key

(year_month, region)

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**route_performance**

| Column | Data Type |
|---|---|
| year | INTEGER |
| route | VARCHAR(100) |
| origin_iata | CHAR(3) |
| destination_iata | CHAR(3) |
| distance_km | INTEGER |
| region | VARCHAR(50) |
| main_airlines | VARCHAR(255) |
| annual_passengers_m | NUMERIC(10,2) |
| avg_fare_usd | NUMERIC(10,2) |
| weekly_frequency_est | INTEGER |
| annual_revenue_usd_m | NUMERIC(12,2) |

Primary Key

(route, year)

---

**aviation_incidents**

| Column | Data Type |
|---|---|
| incident_id | VARCHAR(20) |
| date | DATE |
| year | INTEGER |
| month | INTEGER |
| flight_number | VARCHAR(20) |
| airline | VARCHAR(100) |
| aircraft_type | VARCHAR(100) |
| severity | VARCHAR(30) |
| fatalities | INTEGER |
| location | VARCHAR(150) |
| description | TEXT |
| is_fatal | BOOLEAN |
| is_geopolitical | BOOLEAN |
| is_boeing | BOOLEAN |
| is_airbus | BOOLEAN |

Primary Key

incident_id

---

**Data Import Process**

The CSV files were imported into PostgreSQL using pgAdmin's Import/Export Data feature.

Import sequence:

1. airline_financials.csv
2. fleet_orders.csv
3. passenger_traffic.csv
4. route_performance.csv
5. aviation_incidents.csv

After each import, the table contents were verified using SQL queries to ensure successful loading.

---

**Data Validation**

The following validation checks were performed after import:

- Verified total row count for each table.

- Confirmed all columns were imported with correct data types.
- Checked for import errors.
- Verified primary key values.
- Confirmed no unexpected truncation or missing records.

Expected row counts:

| Table | Expected Rows |
|---|---|
| airline_financials | 497 |
| fleet_orders | 86 |
| passenger_traffic | 1,176 |
| route_performance | 400 |
| aviation_incidents | 40 |

---

**Logical Relationships**

This project combines multiple analytical datasets rather than a transactional database. Therefore, logical relationships are established using common business attributes instead of foreign keys.

| Table | Related Table | Common Field |
|---|---|---|
| airline_financials | aviation_incidents | Airline Name |
| airline_financials | passenger_traffic | Year |
| passenger_traffic | route_performance | Region, Year |
| fleet_orders | airline_financials | Year |
| fleet_orders | aviation_incidents | Year (Analytical Comparison) |

These relationships support cross-table analysis in SQL and Power BI.

---

**Deliverables**

At the end of this phase, the following deliverables are available:

- PostgreSQL database created
- Five tables designed
- Appropriate data types assigned
- Primary keys defined
- CSV files successfully imported
- Import validation completed
- Logical database schema documented

---

**Outcome**

The aviation_project database is now prepared for SQL-based data cleaning, exploratory analysis, advanced analytics, and Power BI dashboard development. This structured database serves as the foundation for all subsequent phases of the project.

----------------------------------- ----------------------------------------- --------------------------

📄 **Step 4 – Data Cleaning & Validation Documentation**

**4.1 Row Count Validation**

**Objective**

Verify that all CSV files were imported successfully into PostgreSQL.

**Validation Performed**

SELECT COUNT(*) FROM airline_financials;

SELECT COUNT(*) FROM fleet_orders;

SELECT COUNT(*) FROM passenger_traffic;

SELECT COUNT(*) FROM route_performance;

SELECT COUNT(*) FROM aviation_incidents;

**Result**

- All tables were imported successfully.
- Row counts matched the source CSV files.

---

**4.2 Duplicate Record Validation**

**Objective**

Identify duplicate business records using natural business keys.

**Validation**

Example:

SELECT airline_name,

    year,

    COUNT(*)

FROM airline_financials

GROUP BY airline_name, year

HAVING COUNT(*) > 1;

**Result**

No duplicate business records were identified.

---

**4.3 Missing Value Validation**

**Objective**

Detect NULL values in important business columns.

**Validation**

Checked all numeric and descriptive columns for missing values.

Example:

SELECT *

FROM airline_financials

WHERE revenue_usd_bn IS NULL;

**Result**

No critical missing values affecting business analysis.

---

**4.4 Remove Leading & Trailing Spaces**

**Objective**

Standardize text values by removing unnecessary spaces.

**Columns Cleaned**

**airline_financials**

- airline_name
- region
- business_model
- country_iso3
- iata_code

**fleet_orders**

- manufacturer
- aircraft_family

**passenger_traffic**

- region

**route_performance**

- route
- origin_iata
- destination_iata
- region
- main_airlines

**aviation_incidents**

- airline
- aircraft_type
- severity
- location

**SQL Function Used**

TRIM()

**Result**

All unnecessary leading and trailing spaces were removed.

---

**4.5 Text Standardization**

**Objective**

Ensure consistent text formatting.

**Functions Used**

UPPER()

LOWER()

INITCAP()

Examples

north america

North America

NORTH AMERICA

Standardized to

North America

Airport and country codes were standardized using uppercase.

Example

hyd → HYD

usa → USA

---

**4.6 Numeric Validation**

**Objective**

Identify impossible or unrealistic numeric values.

**Validations Performed**

Checked

- Revenue
- Operating Margin
- Operating Income
- Passenger Count
- Load Factor
- Fleet Size
- Orders
- Deliveries
- Backlog

- Route Revenue
- Distance
- Fatalities

Example

```
SELECT *
FROM airline_financials
WHERE load_factor_pct > 100
  OR load_factor_pct < 0;
```

**Result**

No invalid numeric values were identified.

Business exceptions such as negative operating income during COVID-19 were retained because they represent real business conditions rather than data errors.

---

**4.7 Business Rule Validation**

**Objective**

Validate business logic.

Examples

- Load Factor must be between 0 and 100.
- Month must be between 1 and 12.
- Distance cannot be zero.
- Revenue cannot be negative.
- Orders should follow business logic.

**Result**

Business rules were successfully validated.

---

**4.8 Distinct Value Validation**

**Objective**

Detect inconsistent text values.

Example

```
SELECT DISTINCT region
FROM airline_financials;
```

Checked

- Region
- Business Model
- Manufacturer
- Aircraft Family

- Airline
- Severity
- Aircraft Type
- Airport Codes

**Result**

Text values were standardized and made consistent.

---

**4.9 Date Validation**

**Objective**

Validate date ranges.

**Validation**

Confirmed

- Years between 2010 and 2026
- Months between 1 and 12
- Valid incident dates

Example

```
SELECT MIN(year),
    MAX(year)
FROM airline_financials;
```

**Result**

Date ranges matched the project scope.

---

**4.10 Data Quality Summary**

| Validation | Status |
|---|---|
| Import Validation | ✅ Completed |
| Duplicate Check | ✅ Completed |
| Missing Values | ✅ Completed |
| Remove Extra Spaces | ✅ Completed |
| Text Standardization | ✅ Completed |
| Numeric Validation | ✅ Completed |
| Business Rule Validation | ✅ Completed |
| Distinct Value Validation | ✅ Completed |
| Date Validation | ✅ Completed |

---

**Key SQL Functions Used**

| Function | Purpose |
|---|---|
| TRIM() | Remove extra spaces |
| UPPER() | Convert to uppercase |
| LOWER() | Convert to lowercase |
| INITCAP() | Capitalize words |
| DISTINCT | Identify unique values |
| COUNT() | Validate row counts |
| MIN() | Find minimum values |
| MAX() | Find maximum values |
| GROUP BY | Detect duplicates |
| HAVING | Filter duplicate groups |
| IS NULL | Detect missing values |

---

**Outcome**

The datasets were successfully cleaned and validated in PostgreSQL. Data quality issues such as whitespace inconsistencies, text formatting differences, invalid numeric values, duplicate records, and date anomalies were addressed or verified. The resulting tables are consistent, reliable, and ready for **Exploratory Data Analysis (EDA)** and **Power BI dashboard development**.

📄 **Step 5 Documentation – Exploratory Data Analysis (EDA)**

**Business Goals**

- Analyze airline financial performance.
- Identify revenue trends over time.
- Measure passenger growth and demand.
- Compare airline business models.
- Evaluate Boeing vs Airbus performance.
- Analyze route profitability.
- Assess aviation safety trends.

---

**SQL Concepts Used**

- SELECT
- WHERE
- ORDER BY
- GROUP BY
- HAVING
- Aggregate Functions

- DISTINCT
- LIMIT

---

**Business Questions Solved**

**Financial Analysis**

- Which airline generated the highest revenue?
- What is the yearly revenue trend?
- Which airline carried the most passengers?
- Which airline has the highest operating margin?
- Compare legacy and low-cost airlines.

---

**Fleet Analysis**

- Boeing vs Airbus yearly orders.
- Deliveries trend.
- Backlog trend.
- Widebody vs Narrowbody demand.

---

**Passenger Analysis**

- Regional passenger growth.
- COVID recovery.
- Load factor comparison.
- Monthly traffic trend.

---

**Route Analysis**

- Highest revenue routes.
- Highest passenger routes.
- Average fare comparison.

---

**Safety Analysis**

- Fatal incidents by year.
- Boeing vs Airbus incidents.
- Geopolitical events.

---

**Key Insights**

- Airline revenue declined sharply in 2020 because of COVID-19.

- Passenger demand recovered at different rates across regions.
- Airbus gained market share while Boeing experienced challenges after the 737 MAX crisis.
- Middle Eastern airlines showed strong growth after the pandemic.
- Several international routes generated significantly higher revenue than others.

---

**Outcome**

Successfully completed exploratory analysis and identified key KPIs to be used in dashboard development.

---

### 📄 Step 6 Documentation – Advanced SQL Analysis

**Objective**

Perform advanced SQL analysis to generate deeper business insights using analytical SQL techniques.

---

**SQL Features Used**

- Common Table Expressions (CTEs)
- Window Functions
- RANK()
- DENSE_RANK()
- ROW_NUMBER()
- LAG()
- LEAD()
- CASE WHEN
- Subqueries
- Views

---

**Business Scenarios Solved**

**Revenue Growth Analysis**

Calculated Year-over-Year revenue growth for each airline.

---

**Airline Ranking**

Ranked airlines based on yearly revenue.

---

**Regional Leaders**

Identified the highest revenue airline in every region.

---

**Running Revenue**

Calculated cumulative revenue over the years.

**Industry Benchmarking**

Compared airline revenue against yearly industry averages.

**Profitability Analysis**

Identified airlines performing above average operating margin.

**Revenue Classification**

Categorized airlines into:

- High Revenue
- Medium Revenue
- Low Revenue

**Route Performance**

Ranked routes by annual revenue.

**Manufacturer Analysis**

Compared Boeing and Airbus yearly performance.

**View Creation**

Created reusable SQL Views for reporting.

**SQL Skills Demonstrated**

- Analytical Queries
- Ranking Functions
- Running Totals
- Time-Series Analysis
- Business Logic
- Performance Optimization

**Business Insights**

- COVID caused the largest decline in airline revenue during the study period.
- Boeing recovered gradually after the 737 MAX crisis but remained behind Airbus in several years.
- Low-cost carriers generally maintained stronger operating margins.
- Some airlines consistently outperformed the industry average.

**Outcome**

Generated advanced analytical reports that can directly support executive decision-making and Power BI dashboard development.

---

📄 **Step 7 Documentation – Excel Dashboard**

**Objective**

Develop an interactive Excel dashboard to visualize aviation industry performance before building the Power BI dashboard.

---

**Data Source**

Cleaned PostgreSQL tables exported to Excel.

---

**Excel Features Used**

- Excel Tables
- Pivot Tables
- Pivot Charts
- Slicers
- Conditional Formatting
- Named Ranges

---

**KPI Cards**

Created KPI cards for:

- Total Revenue
- Total Passengers
- Average Load Factor
- Total Aircraft Orders
- Total Deliveries
- Total Aviation Incidents

---

**Dashboard Visualizations**

**Financial Dashboard**

- Revenue Trend
- Top Airlines
- Operating Margin Comparison

---

**Passenger Dashboard**

- Passenger Growth
- Regional Traffic

- Load Factor Trend

---

**Fleet Dashboard**

- Boeing vs Airbus Orders
- Deliveries Trend
- Manufacturer Comparison

---

**Route Dashboard**

- Top Revenue Routes
- Average Fare Analysis

---

**Safety Dashboard**

- Incident Trend
- Fatalities
- Aircraft Manufacturer Comparison

---

**Interactive Filters**

Added slicers for:

- Year
- Airline
- Region
- Manufacturer
- Business Model

---

**Dashboard Design Principles**

Applied:

- Consistent color theme
- Clear KPI placement
- Interactive filtering
- Simple chart layout
- Executive-friendly design

---

**Business Value**

The dashboard enables users to:

- Monitor airline financial performance.

- Compare aircraft manufacturers.
- Track passenger demand.
- Analyze route profitability.
- Review aviation safety trends.

---

**Outcome**

Developed a fully interactive Excel dashboard that summarizes the aviation industry's performance from 2010 to 2026 and serves as the foundation for the Power BI dashboard.

------------------------------ --------------------------------------------- ------------------------------

📄 **Step 8 Power BI Analysis and Power Query**

**Objective**

Connect the PostgreSQL database to Power BI and transform the aviation dataset using Power Query to create a clean, consistent, and analysis-ready data model.

**Power Query Features Used**

- PostgreSQL Connector
- Data Type Conversion
- Trim & Clean
- Capitalize Each Word
- Remove Duplicates
- Column Quality Check
- Remove Unnecessary Columns
- Create Reference Queries
- Dimension Table Creation

**Transformations Performed**

- Verified all data types.
- Standardized text values.
- Removed duplicate records.
- Validated null values.
- Optimized columns for reporting.
- Created reusable dimension tables.

**Outcome**

The transformed dataset is ready for data modeling and dashboard development in Power BI.

---

📄 **Step 9 Documentation — Power BI Data Modeling**

**Project: Global Aviation Industry Analysis 2010–2026**

**Step 9: Data Modeling & Star Schema**

**1. Objective**

The objective of Step 9 was to create a structured data model in Power BI that connects the five aviation datasets through a common **Date dimension**.

The model was designed using a simplified **Star Schema**, with one central Dim Date table connected to the five analytical tables.

---

**2. Tables Used**

**Analytical Tables**

- Airline Financials
- Fleet Orders
- Passenger Traffic
- Route Performance
- Aviation Incidents

**Dimension Table**

- Dim Date

No separate dimension tables such as Dim Airline, Dim Region, or Dim Manufacturer were used in this version of the project.

---

**3. Dim Date Table**

A dedicated calendar table was created in Power BI to provide a common time dimension for the entire project.

The calendar covers the project period:

**2010–2026**

The Date table contains:

- Date
- Year
- Month Number
- Month Name
- Quarter

The table was also marked as the official **Date Table** in Power BI.

---

**4. Date Columns in Analytical Tables**

Since some analytical tables contained only year-level information, appropriate date columns were created to establish relationships with the Date table.

Date fields were prepared for:

- Airline Financials
- Fleet Orders
- Passenger Traffic
- Route Performance
- Aviation Incidents

This ensured that all tables could participate in time-based analysis.

**5. Relationships**

The Dim Date table was connected to all five analytical tables.

| From Table | From Column | To Table | To Column | Cardinality |
|---|---|---|---|---|
| Dim Date | Date | Airline Financials | Date | 1:* |
| Dim Date | Date | Fleet Orders | Date | 1:* |
| Dim Date | Date | Passenger Traffic | year_month | 1:* |
| Dim Date | Date | Route Performance | Date | 1:* |
| Dim Date | Date | Aviation Incidents | Date | 1:* |

All relationships were configured as:

**One-to-Many (1:*)**

with **single-direction filtering** from the Date table to the analytical tables.

---

**6. Star Schema**

The final model follows a simplified Star Schema.

```
    ┌──────────┐
    │ Dim Date │
    └────┬─────┘
      │
 ┌───────────┼───────────┐
 │  │  │  │  │
 ▼  ▼  ▼  ▼  ▼
┌─────────┐ ┌─────┐ ┌───────┐ ┌──────┐ ┌─────────┐
│ Airline  ││Fleet ││Passenger││Route ││ Aviation │
│Financials││Orders││ Traffic ││Perf. ││ Incidents │
└─────────┘ └─────┘ └───────┘ └──────┘ └─────────┘
```

---

**7. Why the Star Schema Was Used**

The Star Schema provides a simple and organized structure for analytical reporting.

It allows the project to:

- Analyze multiple datasets using a common date dimension.
- Create consistent time-based filters.
- Build interactive dashboards.
- Perform year-over-year comparisons.
- Support DAX time-intelligence calculations.
- Reduce unnecessary relationships between analytical tables.

---

**8. Relationship Design Principles**

The following principles were followed:

**One-to-Many Relationships**

The Date table represents the **one side**, while the analytical tables represent the **many side**.

**Single-Direction Filtering**

Filters flow from:

**Dim Date → Analytical Tables**

This helps avoid unnecessary filter ambiguity.

**No Direct Fact-to-Fact Relationships**

The five analytical tables were not directly connected to one another.

This avoids unnecessary many-to-many relationships and keeps the model easier to understand.

---

**9. Data Model Validation**

The following checks were performed:

- All five analytical tables were connected to Dim Date.
- All relationships use *1: cardinality**.
- Date fields have appropriate data types.
- Relationships are active.
- Cross-filter direction is set to Single.
- No unnecessary fact-to-fact relationships were created.
- The Date table covers the project's 2010–2026 analysis period.

---

**10. Business Use of the Model**

The model allows users to analyze different aspects of the aviation industry using a common time dimension.

For example, selecting **2020** in a Year slicer can help analyze:

- Airline revenue during COVID-19
- Passenger traffic decline
- Aircraft orders and deliveries
- Route performance
- Aviation incidents

This makes the model useful for understanding major industry events and their impact across different aviation datasets.

---

**11. Step 9 Outcome**

At the end of Step 9, the following were completed:

✅ Five analytical tables connected
✅ One central Dim Date table created
✅ Year, month, and quarter fields prepared
✅ Date fields prepared for analytical tables
✅ Five Date relationships created
✅ One-to-many cardinality established
✅ Single-direction filtering configured
✅ Simplified Star Schema completed
✅ Model validated for Power BI reporting

---

**Step 10: DAX Measures & Calculations**

**1. Objective**

The objective of Step 10 was to create **DAX measures in Power BI** for calculating important business KPIs and analytical metrics.

These measures will be reused across the Power BI dashboards to analyze:

- Airline financial performance
- Passenger performance
- Aircraft orders and deliveries
- Route performance
- Aviation safety
- Year-over-year growth
- Time-based performance

---

**2. Measures Table**

A dedicated **Measures** table was created to organize the DAX measures in one location.

This keeps the Power BI model clean and makes the measures easier to find and maintain.

---

**3. Financial Performance Measures**

**Total Revenue**

Calculates the total revenue generated by the airlines.

**Business purpose:**
Used to evaluate overall airline financial performance and compare revenue across years, regions, and airlines.

---

**Revenue Growth %**

Measures the percentage change in revenue compared with the previous year.

**Business purpose:**
Used to identify periods of revenue growth or decline, particularly during major events such as COVID-19.

---

**Revenue YTD**

Calculates cumulative revenue from the beginning of the selected year through the selected date.

**Business purpose:**
Used to monitor revenue progression throughout a year.

---

**4. Passenger Measures**

**Total Passengers**

Calculates the total number of passengers carried by the airlines.

**Business purpose:**
Used to analyze passenger demand and compare airline or regional performance.

---

**Average Load Factor**

Calculates the average percentage of available seats occupied by passengers.

**Business purpose:**
Used to measure airline capacity utilization and operational efficiency.

---

**5. Aircraft & Fleet Measures**

**Total Orders**

Calculates the total net aircraft orders.

**Business purpose:**
Used to compare aircraft demand and manufacturer performance.

---

**Total Deliveries**

Calculates the number of aircraft delivered.

**Business purpose:**
Used to evaluate aircraft production and delivery performance.

---

**Total Backlog**

Calculates the total outstanding aircraft orders.

**Business purpose:**
Used to understand future aircraft demand and manufacturer order pipelines.

---

**6. Passenger Traffic Measures**

**Total RPK**

Calculates total Revenue Passenger Kilometers.

**Business purpose:**
Measures passenger demand based on the distance traveled.

---

**Total ASK**

Calculates total Available Seat Kilometers.

**Business purpose:**
Measures the amount of passenger capacity supplied by airlines.

---

**Load Factor Analysis**

RPK and ASK can be compared to understand how effectively available passenger capacity is being utilized.

**Business purpose:**
Helps identify changes in demand versus available capacity.

---

**7. Route Performance Measures**

**Route Revenue**

Calculates total estimated revenue generated across the analyzed routes.

**Business purpose:**
Used to identify high-value routes and compare route-level performance.

---

**Average Fare**

Calculates the average fare across analyzed routes.

**Business purpose:**
Helps understand pricing differences between routes and markets.

---

**8. Aviation Safety Measures**

**Total Incidents**

Counts the number of aviation incidents included in the dataset.

**Business purpose:**
Used to analyze aviation safety trends over time.

---

**Total Fatalities**

Calculates the total number of fatalities recorded in the aviation incidents dataset.

**Business purpose:**
Used to evaluate the severity and impact of major aviation incidents.

---

**9. Airline Ranking**

**Airline Rank**

Ranks airlines based on their total revenue.

**Business purpose:**
Allows users to identify the largest airlines by revenue and compare their relative positions.

---

**10. Time Intelligence**

The DAX measures use the **Dim Date** table created during Step 9.

This enables analysis such as:

- Year-over-Year Revenue Growth
- Previous-year comparison
- Year-to-Date Revenue
- Historical trends

- COVID-19 period comparison
- Recovery analysis

---

**11. KPI Measures Created**

The following measures were created:

| Measure | Business Purpose |
|---|---|
| Total Revenue | Overall airline revenue |
| Total Passengers | Passenger volume |
| Average Load Factor | Capacity utilization |
| Total Orders | Aircraft demand |
| Total Deliveries | Aircraft production/delivery |
| Total Backlog | Future aircraft demand |
| Total RPK | Passenger demand |
| Total ASK | Passenger capacity |
| Route Revenue | Route-level revenue |
| Average Fare | Route pricing |
| Total Incidents | Number of major incidents |
| Total Fatalities | Severity of incidents |
| Revenue Growth % | Year-over-year performance |
| Revenue YTD | Cumulative annual revenue |
| Airline Rank | Airline revenue ranking |

---

**12. Why DAX Measures Were Used**

DAX measures were used instead of manually calculating values in individual visuals because measures:

- Can be reused across multiple dashboard pages.
- Respond dynamically to slicers and filters.
- Support time-intelligence analysis.
- Keep calculations centralized.
- Improve consistency across reports.
- Allow users to interact with the dashboard dynamically.

For example, when a user selects:

**2020 → Asia Pacific → specific airline**

the KPI measures automatically recalculate according to the selected filters.

**14. Validation**

The measures were tested using different:

- Years
- Airlines
- Regions
- Business models
- Manufacturers
- Dashboard filters

The results were checked against the underlying dataset to ensure that the calculations responded correctly to filter selections.

---

#### 15. Step 10 Outcome

At the end of Step 10, the DAX calculation layer was completed.

**Completed:**

✅ Measures table created
✅ Financial KPIs created
✅ Passenger KPIs created
✅ Aircraft KPIs created
✅ Route KPIs created
✅ Safety KPIs created
✅ Revenue growth calculation created
✅ YTD calculation created
✅ Airline ranking created
✅ Time-intelligence calculations prepared
✅ Measures validated with filters

---

📄 STEP 11 — POWER BI DASHBOARD DEVELOPMENT

Project: Global Aviation Industry Analysis 2010–2026

Objective

The objective of Step 11 was to create an interactive Power BI dashboard using the cleaned and transformed aviation datasets.

The dashboard was designed to analyze:

- Airline financial performance
- Passenger demand
- Aircraft manufacturer competition
- Route performance
- Aviation incidents and fatalities

---

11.1 Dashboard Structure

Five Power BI dashboard pages were created:

| Page | Dashboard | Main Purpose |
|---|---|---|
| Page 1 | Executive Overview | Overall industry performance |

| Page | Dashboard | Main Purpose |
|---|---|---|
| Page 2 | Airline Performance | Airline financial and operational analysis |
| Page 3 | Boeing vs Airbus | Aircraft manufacturer competition |
| Page 4 | Passenger & Route Analysis | Passenger demand and route performance |
| Page 5 | Aviation Safety | Major aviation incidents and fatalities |

---

11.2 Page 1 — Executive Overview

Objective

The Executive Overview provides a high-level view of the global aviation industry's performance from 2010–2026.

KPI Cards

- Total Revenue
- Total Passengers
- Average Load Factor
- Total Orders
- Total Deliveries
- Total Incidents

Visualizations

Revenue Trend

- Line Chart
- Year vs Total Revenue

Passenger Trend

- Area Chart
- Year vs Total Passengers

Revenue by Region

- Bar Chart
- Region vs Total Revenue

Top 10 Airlines

- Bar Chart
- Airline vs Total Revenue

Slicers

- Year
- Region
- Airline
- Business Model

Purpose

This page provides executives with a quick overview of industry financial and operational performance.

---

11.3 Page 2 — Airline Performance

Objective

This page analyzes individual airline financial and operational performance.

KPI Cards

- Total Revenue
- Total Passengers
- Average Load Factor
- Revenue Growth

Visualizations

Revenue by Airline

- Compares revenue generated by airlines.

Operating Margin by Airline

- Compares profitability across airlines.

Passengers by Airline

- Shows passenger volume handled by each airline.

Fleet Size by Airline

- Compares estimated fleet sizes.

Revenue by Business Model

- Compares revenue across airline business models.

Revenue vs Passengers

- Scatter chart comparing passenger volume and revenue.
- Fleet size is used as an additional variable.

Slicers

- Year
- Airline
- Region
- Business Model

Purpose

This page helps identify high-performing airlines, revenue leaders, passenger leaders, and differences between airline business models.

---

11.4 Page 3 — Boeing vs Airbus Analysis

Objective

This page analyzes aircraft manufacturer competition between:

- Boeing
- Airbus
- COMAC
- Embraer

KPI Cards

- Total Net Orders
- Total Deliveries
- Total Backlog
- Number of Manufacturers

Visualizations

Aircraft Orders by Manufacturer

- Compares annual orders between manufacturers.

Aircraft Deliveries by Manufacturer

- Compares aircraft deliveries over time.

Backlog by Manufacturer

- Compares outstanding aircraft orders.

Orders by Aircraft Family

- Identifies aircraft families with the highest demand.

Boeing vs Airbus Order Trend

- Focuses on the competitive performance of Boeing and Airbus.

Orders by Aircraft Category

- Compares:
  - Narrowbody
  - Widebody
  - Regional
  - Other

Slicers

- Manufacturer
- Year

Purpose

This page helps evaluate aircraft manufacturer competition, aircraft demand, backlog, and the development of COMAC.

---

11.5 Page 4 — Passenger & Route Analysis

Objective

This page analyzes passenger demand, regional traffic, and global route performance.

KPI Cards

- Total RPK
- Total ASK
- Average Load Factor
- Route Revenue
- Average Fare

Visualizations

Regional Passenger Traffic Trend

- Shows passenger demand by region over time.

Regional Load Factor Trend

- Compares passenger load factors between regions.

Passenger Traffic by Region

- Compares total passenger demand across regions.

Top 10 Global Routes

- Identifies routes with the highest passenger volume.

Top 10 Routes by Revenue

- Identifies routes with the highest estimated revenue.

Route Fare vs Passenger Volume

- Scatter chart comparing average fare with passenger volume.
- Route revenue is used as an additional variable.

Slicers

- Region
- Year
- Route

Purpose

This page supports analysis of COVID-19 impact, regional recovery, route demand, route revenue, and pricing.

---

11.6 Page 5 — Aviation Safety

Objective

This page analyzes the major aviation incidents included in the dataset.

Important: The dataset contains selected major incidents, not every aviation incident worldwide. Therefore, the dashboard is used for analysis of the selected incident dataset, rather than as a complete global aviation safety measurement.

KPI Cards

- Total Incidents
- Total Fatalities
- Fatal Incidents
- Geopolitical Incidents

Visualizations

Incidents by Year

- Shows the number of selected major incidents by year.

Fatalities by Year

- Shows annual fatalities.

Incidents by Severity

- Categorizes incidents by severity.

Categories include:

- Fatal Crash
- Fatal Collision
- Incident
- Ground Incident
- Sanctions Impact

Boeing vs Airbus Incidents

- Compares selected incidents involving Boeing and Airbus aircraft.

Incidents by Airline

- Shows the number of selected incidents by airline.

Fatalities by Aircraft Type

- Compares fatalities across aircraft types.

Geopolitical vs Operational Incidents

- Separates geopolitical incidents from other incidents.

Slicers

- Year
- Severity

---

11.7 Data Model

The Power BI model uses a simplified Star Schema.

The Dim Date table acts as the central date dimension.

Dim Date

```
                 │
      ┌──────────────────────────────┼──────────────────────────────┐
      │          │          │
      ▼          ▼          ▼
 Airline Financials  Fleet Orders   Passenger Traffic
      │
      │
      ▼
 Route Performance

      │
      ▼
 Aviation Incidents
```

Relationships were created between:

Dim Date → Analytical Tables

with:

Cardinality: 1-to-many (1:*)

The Dim Date table allows users to analyze the different datasets consistently by year and date.

---

11.8 Interactive Features

Interactive filtering was added throughout the dashboard.

Filters Used

- Year
- Region
- Airline
- Business Model
- Manufacturer
- Route
- Severity

These filters allow users to drill down from the overall industry level into specific airlines, regions, manufacturers, routes, and time periods.

---

11.9 Dashboard Design

The dashboards were designed using:

- KPI Cards

- Bar Charts
- Column Charts
- Line Charts
- Area Charts
- Donut Charts
- Scatter Charts
- Slicers

The visual layout was organized to place KPIs at the top, followed by trend analysis and detailed comparison visuals.

---

11.10 Business Questions Addressed

The dashboards were designed to answer important business questions such as:

1. How has aviation industry revenue changed from 2010–2026?
2. How did COVID-19 affect airline revenue and passenger traffic?
3. Which airlines generated the highest revenue?
4. Which business models performed better?
5. Which regions recovered fastest after COVID-19?
6. How did Boeing's orders change during the 737 MAX crisis?
7. How does Boeing compare with Airbus in orders and backlog?
8. Is COMAC becoming a significant competitor?
9. Which global routes have the highest passenger volumes?
10. Which routes generate the highest estimated revenue?
11. What is the relationship between fares and passenger volume?
12. How have major incidents changed over time?
13. Which years recorded the highest fatalities?
14. How do selected Boeing and Airbus incidents compare?
15. What role do geopolitical events play in the selected incidents?

---

11.11 Tools Used

| Tool | Purpose |
|---|---|
| PostgreSQL | Data storage and SQL analysis |
| Excel | Data validation and preliminary analysis |
| Power Query | Data cleaning and transformation |
| Power BI | Data modeling and dashboard development |
| DAX | Measures and analytical calculations |

11.12 Final Outcome

At the end of Step 11, a five-page interactive Power BI dashboard was successfully developed.

The dashboard provides a comprehensive view of:

Airline Financials ➔ Airline Performance ➔ Aircraft Manufacturers ➔ Passenger Traffic ➔ Route Economics ➔ Aviation Safety

The dashboard can be used by business stakeholders to identify trends, compare performance, investigate disruptions, and support strategic decision-making.

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